Introduction to SQL Databases (2025)

A three-day foundation in relational databases, built on SQL Server 2025. It gives programmers, system administrators and IT professionals the design, T-SQL and administration skills to start working with data.

DURATION	CONTENT	LEVEL	DELIVERY
3 Days	9 Modules	Beginner	Instructor-Led · Bootcamp · Self-Paced

WHAT SETS THIS COURSE APART

01

Updated for SQL Server 2025

Demonstrations and hands-on labs run on SQL Server 2025 with Management Studio (SSMS) 22.

- SQL Server 2025 architecture taught from Module 1
- Optional module on what is new in the release
- Enhanced temporal tables, security, performance and JSON processing

02

AI features in the database engine

Students learn how SQL Server 2025 supports AI workloads through its vector capabilities.

- Vector databases and their use cases for AI workloads
- Hands-on lab on vector search
- Vector similarity queries in the SELECT tutorial

03

Lab simulations, at the student's pace

Students step back and forth through each lab exercise, revisiting any step as needed.

- Runs in a browser, with no setup required
- Replay any time, including after class
- Covers the lab in all 9 modules

ALSO INCLUDED

AI Tutor, 24/7	An interactive AI assistant built into the courseware platform answers questions on every course topic, at no extra cost.
Bootcamp mode	One capstone lab per day. Students apply that day's skills to a new scenario with no procedure and no hints.
PowerShell	Scripted administration and backups using the SqlServer PowerShell module.
Performance	Indexing strategies and execution plan analysis to find and fix bottlenecks.

CURRICULUM AT A GLANCE

DAY 1	**Foundations.** Introduction to databases, database languages (T-SQL), and data modeling techniques.
DAY 2	**Design.** Normalization and denormalization, relationship types, and the effects of design on performance.
DAY 3	**Build and automate.** Views, stored procedures, functions and triggers; PowerShell for SQL Server; what is new in SQL Server 2025 (optional).

What your team leaves with

The ability to design normalized database schemas, write T-SQL to create and query database objects, enforce data integrity, tune queries with indexes, automate administration with PowerShell, and identify and apply the new features in SQL Server 2025.

Who it is for: programmers, system administrators and IT professionals expanding into database management.
Prerequisites: basic Windows and general IT knowledge. No prior database experience required.

COURSEWARE ACCESS

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